

July 2021 Term
CSE 471 – Machine Learning
Class Test# 2 Time: 20 minutes

Name: _____ Student No. _____

No extra sheet shall be provided.

1. Consider the Bayesian Network (BN) structure along with the various survey results as specified below.

- In a survey, 1% respondents selected “YES” to having encountered a significant fall at any point during their life.
- The global diabetes prevalence in 2019 is estimated to be 9.3%.
- Table 1 summarizes another survey result on Diabetic and Non-diabetic people. The number of data points is mentioned directly in a cell, or inside parenthesis, as appropriate. As an example, from Table 1, we can see that 30 out of 100 diabetic patients have hypothyroidism. Of 50 patients having diabetes as well as a fall event in their life, only 20 had disk prolapse.
- Table 2 gives the CPT of back pain event.

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graph TD; D((Diabetes D)) --> S((Spondylitis S)); D((Diabetes D)) --> DP((Disk prolapse DP)); F((Fall F)) --> DP((Disk prolapse DP)); S((Spondylitis S)) --> B((Back pain B)); DP((Disk prolapse DP)) --> B((Back pain B)); D((Diabetes D)) --> H((Hypothyroidism H))
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Table 1: Results of a survey

	Hypothyroidism	Spondylitis	Disk prolapse	
Diabetic (100)	30	25	20	Fall (50)
			12	No fall (50)
Non-diabetic (100)	10	15	10	Fall (50)
			4	No fall (50)

Table 2: CPT of Back pain

S	DP	P(B=true S, DP)
t	t	0.90
t	f	0.65
f	t	0.75
f	f	0.30

Based on the above information, please do the following.

1. Construct the CPT table for each node of the network. (10)
2. Calculate the posterior probability of a person having diabetes, given s/he never had a fall, has back pain but no hypothyroidism. (10)